# The Bi-Infinite Series Number System
# A Unified Additive Real–$p$-Adic Structure (Paper II)

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**Abstract**

This Paper II develops the algebraic foundations of the bi-infinite series number system introduced in Paper I. We formalize the evaluation of bi-infinite digit strings by unifying real and $p$-adic interpretations. We show that this unified evaluation yields a valid additive number system. However, we prove that multiplication cannot be defined globally due to two-directional infinite carries and divergent convolution sums. The resulting object is a legitimate algebraic structure that exhibits collapse phenomena and extends classical number systems.

## 1 Introduction (Relation to Paper I)

Paper I introduced the idea of representing numbers using bi-infinite digit strings of the form

$$\dots d_2 d_1 d_0 . d_{-1} d_{-2} d_{-3} \dots,$$

along with the surprising "collapse" of the symmetric maximal-digit string:

$$\dots (p-1)(p-1)(p-1).(p-1)(p-1)(p-1)\dots \mapsto 0.$$

This follow-up paper presents a complete algebraic treatment, defining the underlying number system and proving its additive consistency while explaining rigorously why multiplication cannot exist.

## 2 Motivation and Overview

Real numbers allow infinite digits to the right of the decimal point, while $p$-adic numbers allow infinite digits to the left. A bi-infinite representation merges both into one framework. The key idea is to interpret the left half $p$-adically and the right half using real-limit semantics. The challenge: unify them into a single algebraic object.

## 3 Preliminaries

Fix a prime $p$. Let $\mathcal{D}_p = \{0, 1, \dots, p-1\}$. A $p$-adic integer has the expansion

$$x = \sum_{k=0}^{\infty} a_k p^k.$$

A real fractional expansion is

$$y = \sum_{k=1}^{\infty} b_k p^{-k}.$$

These rely on two different metrics.

## 4 Formal Definitions

**Definition 1** (Bi-infinite string). *A map $d : \mathbb{Z} \to \mathcal{D}_p$ written*

$$\cdots d_2 d_1 d_0 . d_{-1} d_{-2} d_{-3} \cdots .$$

*Let $\Sigma_p = \mathcal{D}_p^{\mathbb{Z}}$.*

**Definition 2** (Left evaluation).

$$L(d) = \sum_{k=0}^{\infty} d_k p^k \in \mathbb{Z}_p.$$

**Definition 3** (Right evaluation).

$$R(d) = \sum_{k=1}^{\infty} d_{-k} p^{-k} \in \mathbb{R}.$$

**Definition 4** (Hybrid value group).

$$\mathbb{V}_p = (\mathbb{Z}_p \oplus \mathbb{R}) / \{(n, -n) : n \in \mathbb{Z}\}.$$

*Write $[x, y]$ for the class.*

**Definition 5** (Unified evaluation).

$$E(d) = [L(d), R(d)].$$

## 5 Main Theorems

**Definition 6** (Maximal symmetric string). *$d_k = p - 1$ for all $k$. Denote this by $\omega$.*

**Lemma 1.** *$L(\omega) = -1$ in $\mathbb{Z}_p$ and $R(\omega) = 1$ in $\mathbb{R}$.*

*Proof.* Right side is geometric. Left side uses $p^{N+1} - 1 \to -1$ $p$-adically. □

**Theorem 1** (Collapse).

$$E(\omega) = 0 \in \mathbb{V}_p.$$

**Theorem 2** (Constant-digit collapse). *For any $d \in \mathcal{D}_p$, the constant string*

$$\cdots dddd.dddd \cdots$$

*evaluates to* 0.

## 6 Negative Numbers without Minus Sign

Any integer $-n$ may be encoded $p$-adically on the left and by $(p - 1)$ on the right.

## 7 Why Multiplication Cannot Exist

Real multiplication works because left digits are finite. $p$-adic multiplication works because right digits are absent, and carries propagate only leftwards. In a bi-infinite system, computing a digit of the product requires

$$c_k = \sum_{i+j=k} d_i e_j,$$

but there are infinitely many $(i, j)$ pairs on both sides. Carries propagate endlessly in both directions, preventing stabilization. Thus multiplication cannot descend to $\mathbb{V}_p$.

## 8 Algebraic Status of the System

The hybrid value group $\mathbb{V}_p$ forms:

- a well-defined abelian group under addition;
- a valid additive number system;
- not a ring or field.

## 9 Future Work

- Restricted multiplication on finite-tail strings.
- Topology and metric geometry of $\mathbb{V}_p$.
- Dynamical systems from shift maps.

## 10 Conclusion

We present a consistent additive extension of real and $p$-adic expansions using bi-infinite strings. The system is algebraically sound under addition and explains collapse behaviors, but cannot support multiplication.

## References


[1] Gouvêa, *p-Adic Numbers.*

[2] Rudin, *Principles of Mathematical Analysis.*